

1. Background and Motivation

- The redistribution of loss and damage finance is debated globally, especially in how support should reach develop-ing countries.
- Empirical evidence on how developed countries behave in these settings is still limited.
- This study uses LLM-agent public-goods games to pre-dict the outcomes from repeated interaction and institu-tional design.

Research Question

- Do developed countries sustain cooperation when con-tributing to a Loss and Damage Fund?
- Does the Fund effectively redistribute resources to devel-oping countries during climate crises?

2. Introduction

This study models repeated climate cooperation with en-dogenous rule changes and budget-constrained contribu-tions. We evaluate whether institutional design can sus-tain mitigation efforts while providing reliable cross-country support after climate shocks. In one-shot play, cooperation is difficult to sustain because agents have incentives to under-contribute. In LDF mode, Stage 1 contributions are treasury-constrained (agent-specific wealth budgets), not fixed equal endowments.

Main Contribution

- 26-country repeated climate public-goods game with treasury-constrained contributions and heterogeneous fis-cal capacity.
- Endogenous governance: periodic constitutional voting updates enforcement and LDF policy parameters.
- Joint economic evaluation of mitigation, welfare, inequal-ity, and LDF transfer efficiency.

3. Literature

- Ostrom (1990)**: Governing the Commons: The Evolution of Institutions for Collective Action.
- Fehr and Gächter (2000)**: Cooperation and Punishment in Public Goods.
- Piedrahita et al. (2025)**: Corrupted by Reasoning: Rea-soning Language Models Become Free-Riders in Public Goods Games.
- Willis et al. (2026)**: Evaluating Collective Behaviour of Hundreds of LLM Agents.
- Tewolde et al. (2026)**: CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas.

4. Design and Identification

Simulation Design

Agents	26 Llama 3.1 7B agents (12 developed, 14 developing)
Rounds	15 repeated rounds
Stage 1	Treasury-based budget cap $0 \leq c_i^{(t)} \leq W_i^{(t)}$, multiplier $\alpha = 1.6$
Initial treasuries	Developed: heterogeneous profile (245, 100, 51, ..., 2); Developing: 3.0 each
Institutions	SI (sanction/reward enabled) and SFI (no sanction channel)
Climate shocks	Realized in rounds 9 and 13
LDF mecha-nism	Pool tracked each round; contributions follow replenishment policy; payouts are damage/vulnerability weighted and coverage-capped
Governance	Constitutional update rounds at 5, 10, and 15

Identification Logic

- Trace mechanism interactions over 15 rounds under a fixed LDF policy architecture.
- Report level, distribution, and transfer metrics jointly to avoid mean-only inference.
- Treat results as within-run mechanism evidence, not cross-design causal identification.
- Use this poster for core evidence and the report for derivations and robustness checks.

Outcome Variables

Contribution path, contribution share of wealth, sanction in-tensity, welfare, inequality, and LDF transfer efficiency.

Hypotheses

- H_1 : Repeated interaction and enforcement can sustain high mitigation effort.
- H_2 : Shock realization raises contribution effort through salience and loss exposure.
- H_3 : LDF payout rules can leave fund inflows much higher than payouts.

5. Theory and Mechanism

Stage 1 — Contributions / Utility

$$\pi_i^{(1,t)} = (W_i^{(t)} - c_i^{(t)}) + \frac{\alpha \sum_{j=1}^n c_j^{(t)}}{n}$$

LDF-mode budget constraint: $0 \leq c_i^{(t)} \leq W_i^{(t)}$; $\text{MPCR} = \alpha/n < 1$ for $n > 1.6$.

Stage 2 — Sanctions / Enforcement

$$\pi_i^{(2)} = E_2 - \sum_j p_{ij} - \sum_j r_{ij} + \beta \sum_j p_{ji} + \rho \sum_j r_{ji}$$

$E_2 = 20$ (SI only), punishment impact $\beta = 3$, reward gain $\rho = 1$.

Subsidy Redistribution

$$\sigma_i = \mathbf{1}\{i \in \mathcal{T}\} \frac{\phi \sum_k \sum_j p_{kj}}{|\mathcal{T}|}$$

Fraction $\phi = 0.2$ of punishment spending is rebated to top con-tributors.

Round Utility

$$U_i^{(t)} = \pi_i^{(1,t)} + \pi_i^{(2,t)} + \sigma_i^{(t)}$$

Cumulative utility captures institutional path dependence.

Mechanism Interpretation

- $\alpha/n < 1$ implies private under-provision despite social gains.
- Deterrence works when expected sanction loss exceeds one-shot deviation gain.
- Rewards and rebates partly offset enforcement burdens for contributors.
- Voting endogenizes constitutional parameters, turning play into institutional adaptation.

Comparative-Statics Predictions

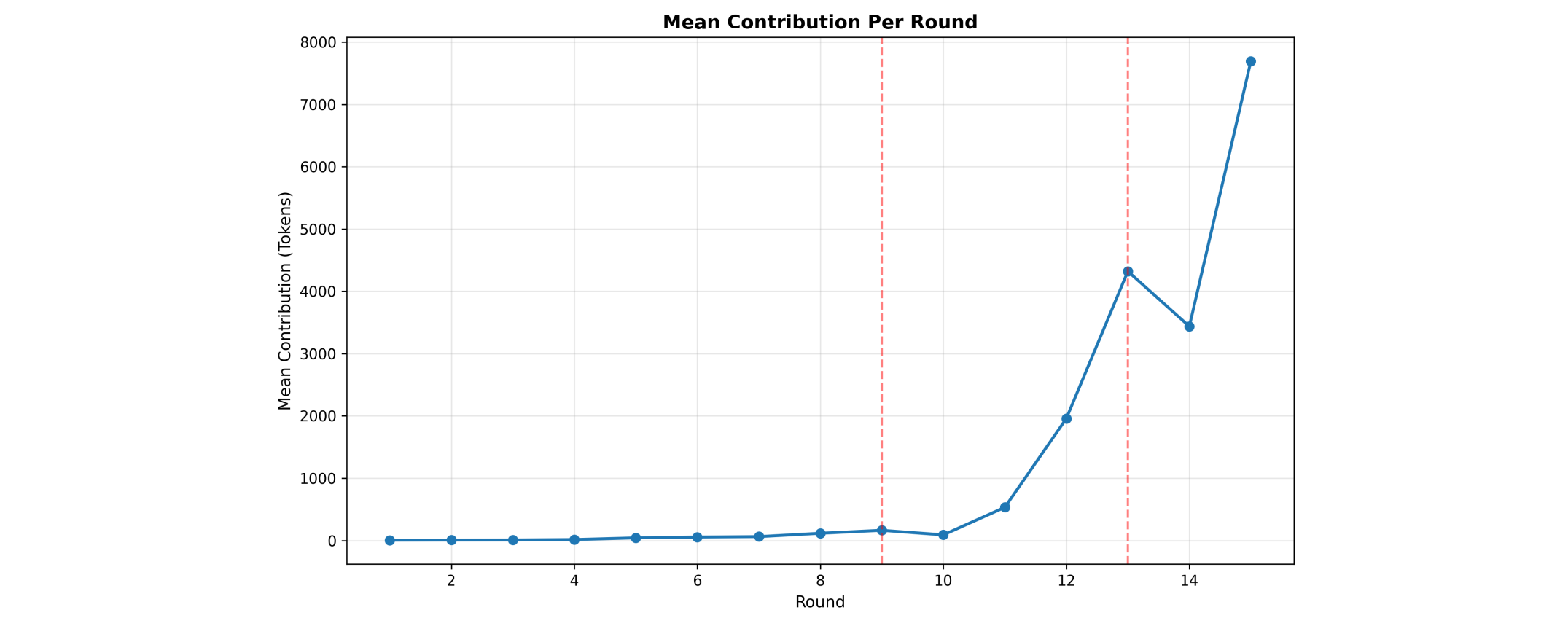
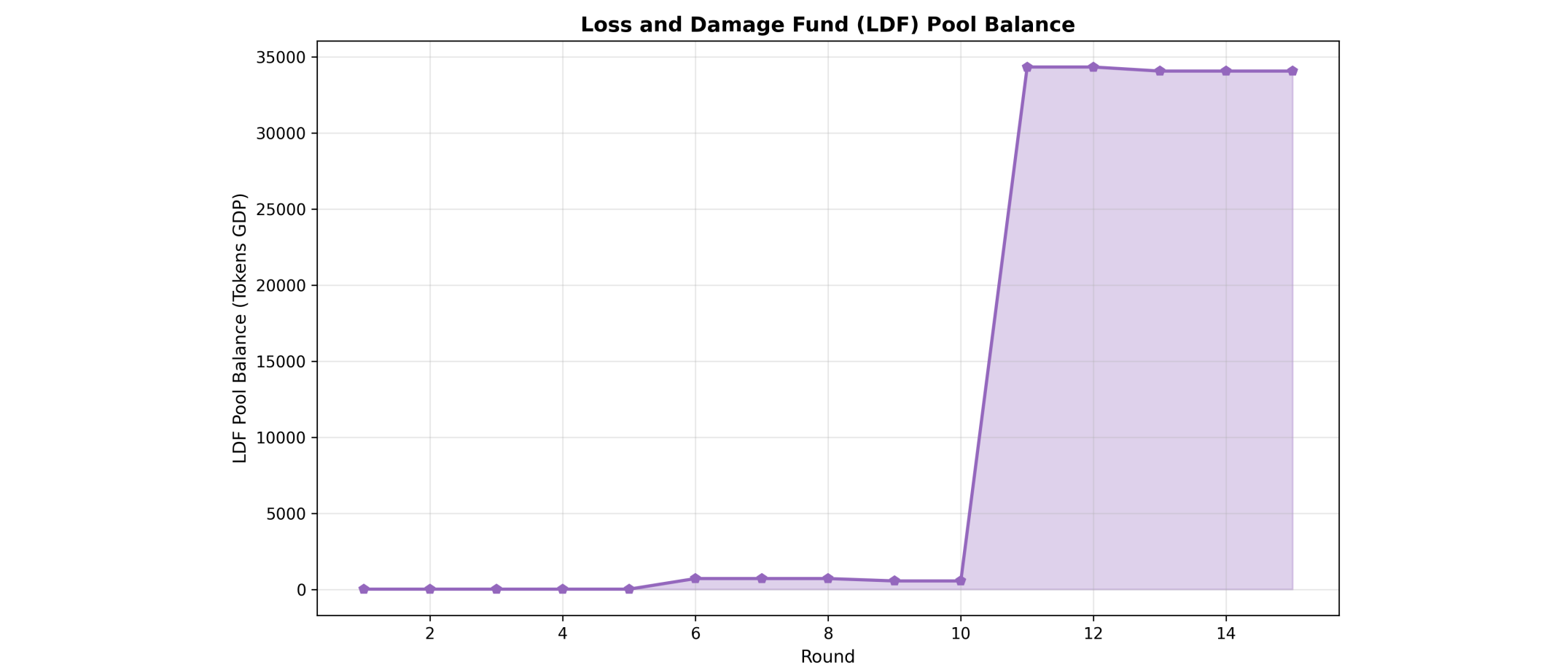
- Larger punishment impact should strengthen deterrence when free-riding pressure rises.
- Higher shock salience should increase contribution effort and accelerate coordination.
- Weak payout rules should increase LDF accumulation while reducing effective insurance value.

6. Results and Policy Implications

Key Results

Metric	Value
Agents / Rounds	26 / 15
Total contributions	481,300
Mean contribution share of wealth	8.1%
Total welfare	18,095,058.63
Sanction intensity	1,686 punish; 1,177 reward
Climate shocks	2 shocks (R9, R13), 471.31 gross damage
LDF contributions vs payouts	34,500.37 vs 424.18
LDF final pool	34,076.18
Mean Gini wealth	0.645

Key Figures



Key Findings

- Mitigation effort scales sharply**: Mean contribution rises from 5.54 to 7,696.08 million dollars (GDP) between rounds 1 and 15, with the steepest acceleration after shock rounds 9 and 13.
- Enforcement is economically active**: SI records 1,686 punishment tokens and 1,177 reward tokens over 15 rounds, indicating non-trivial enforcement activity.
- Redistribution is the binding margin**: 34,500.37 mil-lion dollars (GDP) enters the fund, but only 424.18 is dis-bursed (1.2%), leaving a terminal pool of 34,076.18 and showing that collection vastly outpaces transfer.
- Pooling does not automatically equalize out-comes**: Mean Gini remains 0.645 and the developed-to-developing final wealth ratio stays near 5.55x, so the fund accumulates faster than it redistributes.

Interpretation

- Agents increase contributions after shocks, showing at-tention to losses.
- There is a lot of enforcement activity alongside high giv-ing.
- Most money collected stays in the fund; few payouts hap-pen.
- The fund acts like a savings buffer, not an active payer here.

Conclusion

Repeated interaction and enforcement can raise coopera-tion, but the fund only helps if payouts are designed to be used rather than hoarded.

Policy Implications

- Link incentives to available treasury so poorer agents are not excluded.
- Design payouts and coverage so the fund is actually used.
- Add automatic payout rules or triggers to avoid hoarding.

Limitations

- Institutional assignment is exogenous**: In LDF mode, developed agents are routed to SI and developing agents to SFI in code, so the poster focuses on mechanism per-formance rather than institution-choice behavior.
- Single-run evidence**: This poster highlights one focal re-alization; multi-seed runs would be needed to add uncer-tainty bands around the magnitudes.
- Shock process is exogenous**: Climate shocks are parameter-driven stochastic events, so the results de-scribe adaptation under realized shock timing and sever-ity.